

API Exploration Report

Day 7 — API Fundamentals & Postman | Open-Meteo Weather API

Overview

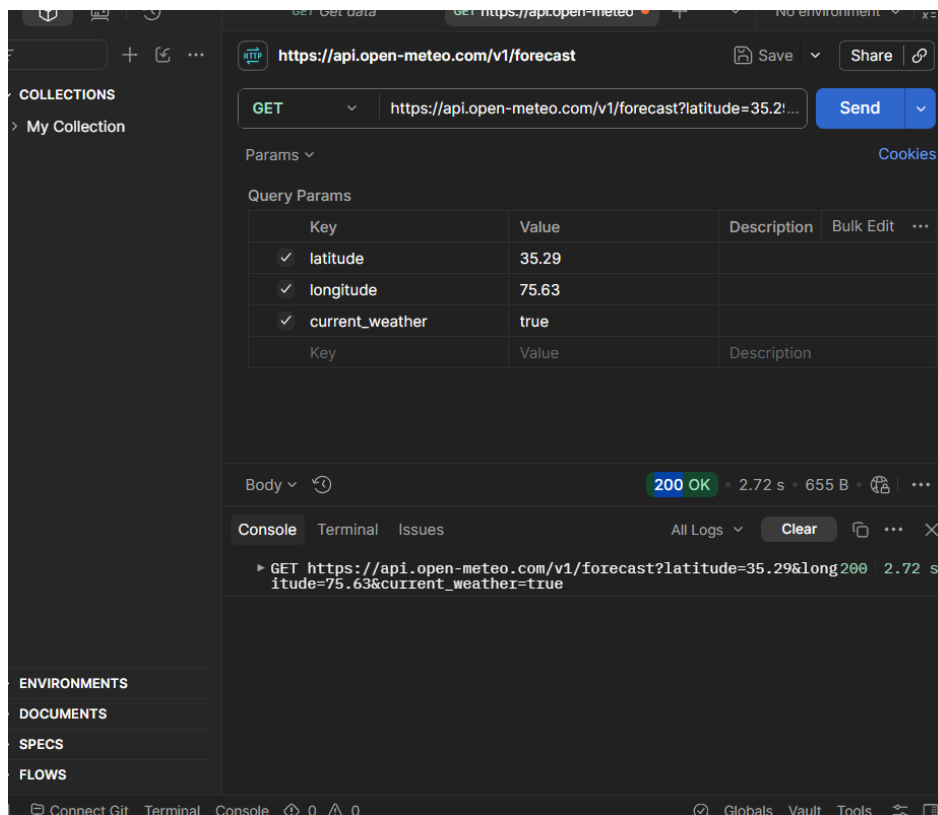
This report documents a hands-on exploration of a free, public REST API using Postman. The Open-Meteo Weather API was chosen because it requires no signup or API key, making it ideal for practicing the fundamentals of sending requests and reading responses.

1. Request Details

A GET request was sent to the Open-Meteo forecast endpoint, requesting current weather data for a location near Skardu, Gilgit-Baltistan.

Method	GET
Endpoint	https://api.open-meteo.com/v1/forecast
Query Parameters	latitude = 35.29 longitude = 75.63 current_weather = true
Full URL	https://api.open-meteo.com/v1/forecast?latitude=35.29&longitude=75.63¤t_weather=true

Screenshot — Request configuration in Postman:



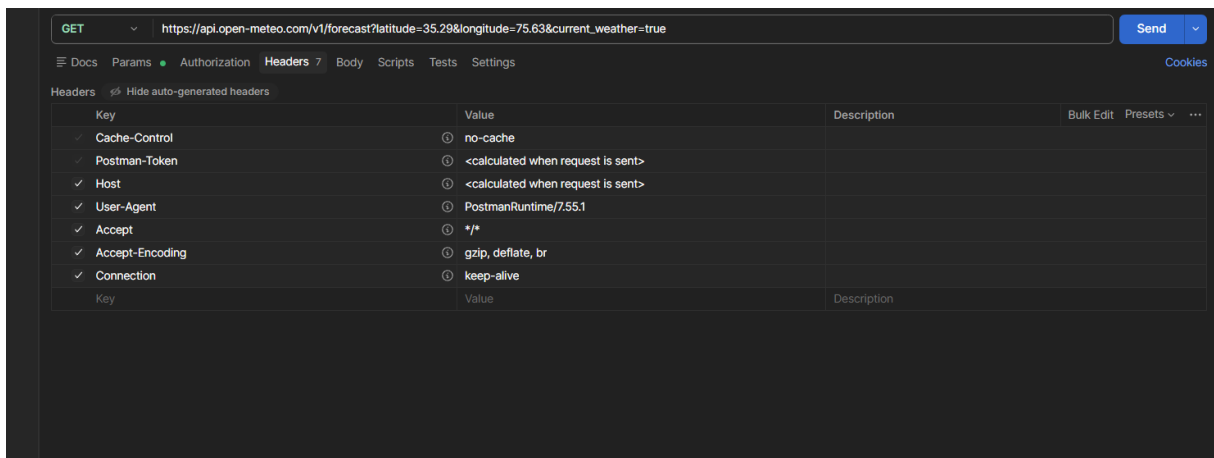
The Params tab shows the three query parameters (*latitude*, *longitude*, *current_weather*) as key/value pairs, which Postman automatically appends to the URL.

2. Request Headers

No custom headers were required since this API is open and doesn't need authentication. Postman auto-generates a standard set of headers for every request, shown below.

Header	Value
Cache-Control	no-cache
Host	<calculated when request is sent>
User-Agent	PostmanRuntime/7.55.1
Accept	*/*
Accept-Encoding	gzip, deflate, br
Connection	keep-alive

Screenshot — Auto-generated request headers in Postman:



Clicking "7 hidden" / "Show auto-generated headers" reveals headers Postman sends by default on every request, even without manual configuration.

3. Successful Response (200 OK)

Status Code	200 OK
Response Time	2.72 s
Response Size	655 B

Screenshot — Response body (JSON):

```
{ } JSON  Preview  Visualize  v
1  {
2    "latitude": 35.32513,
3    "longitude": 75.61381,
4    "generationtime_ms": 6.562232971191406,
5    "utc_offset_seconds": 0,
6    "timezone": "GMT",
7    "timezone_abbreviation": "GMT",
8    "elevation": 2241.0,
9    "current_weather_units": {
10     "time": "iso8601",
11     "interval": "seconds",
12     "temperature": "°C",
13     "windspeed": "km/h",
14     "winddirection": "°",
15     "is_day": " ",
16     "weathercode": "wmo code"
17   },
18   "current_weather": {
19     "time": "2026-07-25T06:15",
20     "interval": 900,
21     "temperature": 19.3,
22     "windspeed": 2.2,
23     "winddirection": 55,
24     "is_day": 1,
25     "weathercode": 51
26   }
27 }
```

The full JSON response returned by the API for the current weather request.

Response Structure Notes

The response is a single JSON object with two layers:

Top-level fields — location and metadata: latitude, longitude, elevation, generationtime_ms, timezone, timezone_abbreviation, utc_offset_seconds.

current_weather_units (nested object) — defines the unit for each measurement below, e.g. temperature in °C, windspeed in km/h, winddirection in °.

current_weather (nested object) — the actual current values: time, interval, temperature (19.3°C), windspeed (2.2 km/h), winddirection (55°), is_day, and weathercode (51).

This is a common REST API pattern: pairing a data object with a matching "units" object so the client can label values correctly without guessing.

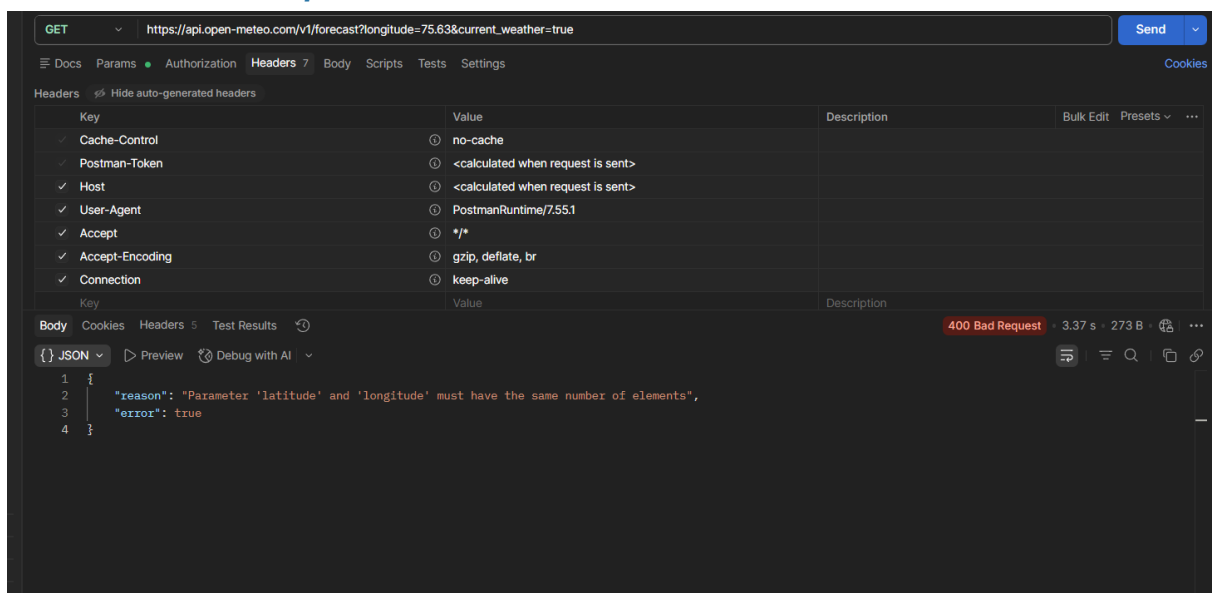
4. Error Response (400 Bad Request)

To see how the API handles invalid input, the request was deliberately broken by removing the **latitude** parameter while keeping **longitude**.

```
https://api.open-meteo.com/v1/forecast?longitude=75.63&current_weather=true
```

Status Code	400 Bad Request
Response Time	3.37 s
Response Size	273 B

Screenshot — Error response in Postman:



The API returned a 400 status with a JSON body explaining the problem.

Why This Happened

The API's error message states: *"Parameter 'latitude' and 'longitude' must have the same number of elements."* Open-Meteo expects latitude and longitude to always be supplied together as a matched pair (it even supports multiple comma-separated coordinate pairs for batch queries). Submitting only one of the two left the pair mismatched, so the server rejected the request before doing any weather lookup — a client-side mistake, correctly reported as a 4xx error rather than a 5xx server error.

5. Key Takeaways

- A REST API request is defined by a method (GET), an endpoint URL, and optional query parameters.
- Even "no-auth-needed" APIs still send a standard set of request headers automatically — Host, User-Agent, Accept, etc.

- A successful response (200 OK) returns structured JSON, often pairing data with a units/metadata object.
- Invalid input produces a 4xx client error, and well-designed APIs return a descriptive JSON error message explaining exactly what went wrong — in this case, a missing required parameter.